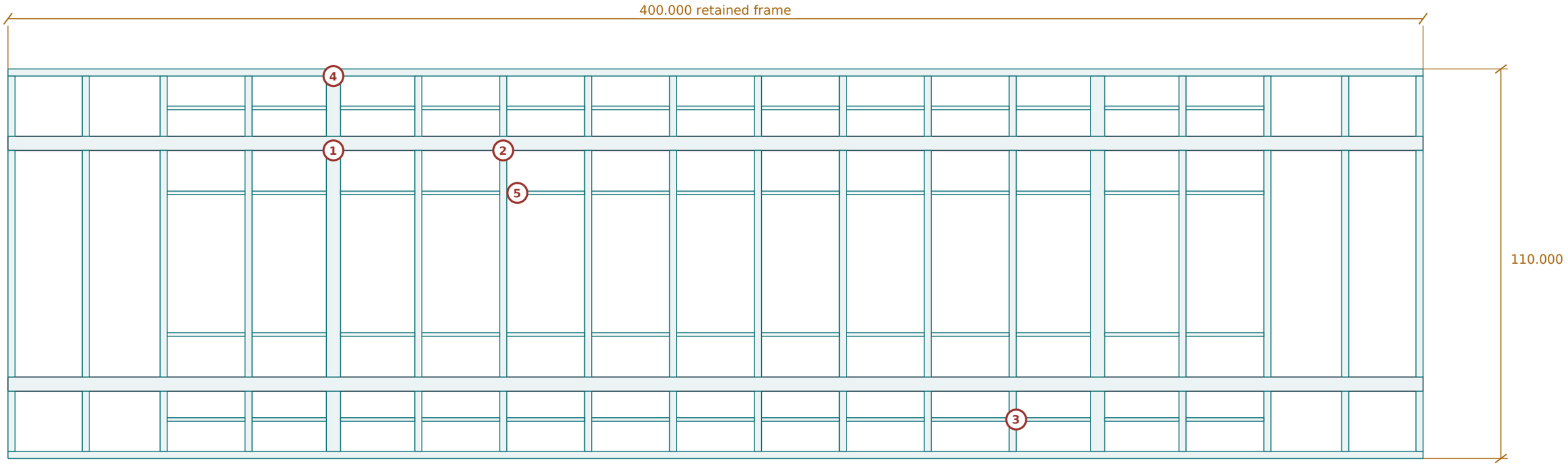
CONNECTION DEVELOPMENT / RETAINED INTERFACES

I1 connection options are separate from the unchanged 110-member F1 frame. All stated physical dimensions are engineering-study geometry.

I1.1 FRAME REFERENCE PLAN / ACTUAL COORDINATE BASIS



Reference labels: 1 bolster/main; 2 ordinary crossmember; 3 seat-rail end; 4 bolster/perimeter; 5 seat hardpoint.

x = front inside end wall toward rear. y = vehicle right. z = top of rail. Forward direction = -x.

I1.2 RETAINED DATUMS / INCHES

Datum	Nominal coordinate
A / steel top	31.003937
Finished floor / F1-E1 study	32.000000
Main girder y centres	-34.000 / +34.000
Bolster / bogie x references	45.000 / 261.000
Seat rails y centres	-44 / -20 / +20 / +44

I1.3 WHAT THIS REVISION CAN ESTABLISH

Local dimensions, calculated demands, trial reinforcement and assembly access can now be reviewed together. A drawn plate or tube is not a released component. Accepted vehicle load cases, fatigue life, material toughness, weld procedures and specialist approval remain controlling.

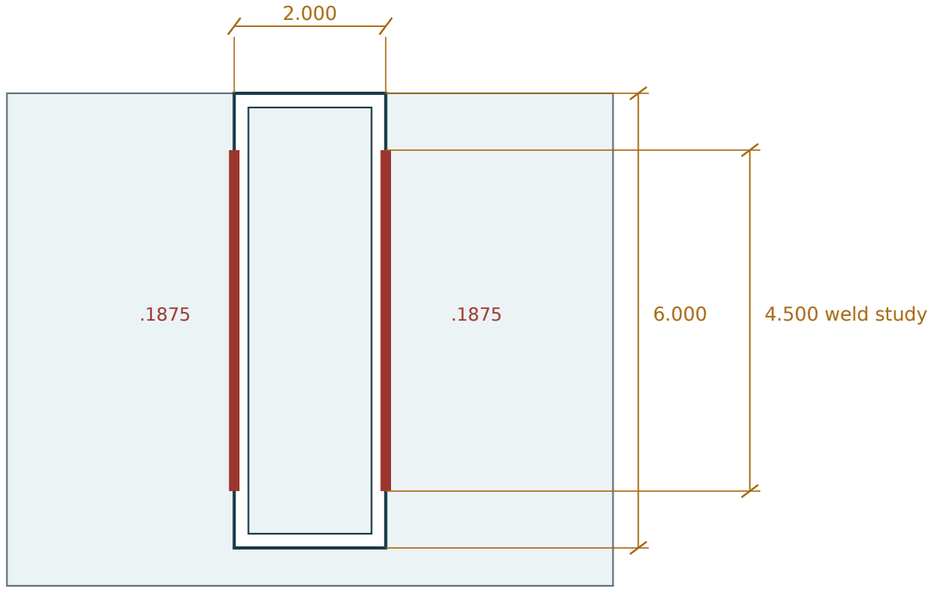
0% track grade remains the owner target. It does not remove vertical dynamics, curving, emergency braking or seat-anchor loads. Bogie/coupler/air-brake interfaces retain their separate engineering holds.

VERIFIED / source data CALCULATED / stated inputs STUDY / proposed geometry HELD / required before release

I-CM6 / DEEPER ORDINARY CROSSMEMBER CONNECTION

Proposed local change: sixteen 64 in central spans become HSS 6 x 2 x 3/16. Outer arms and the original F1 file are unchanged.

I2.1 CONNECTION ELEVATION / x-z / AT y = +32

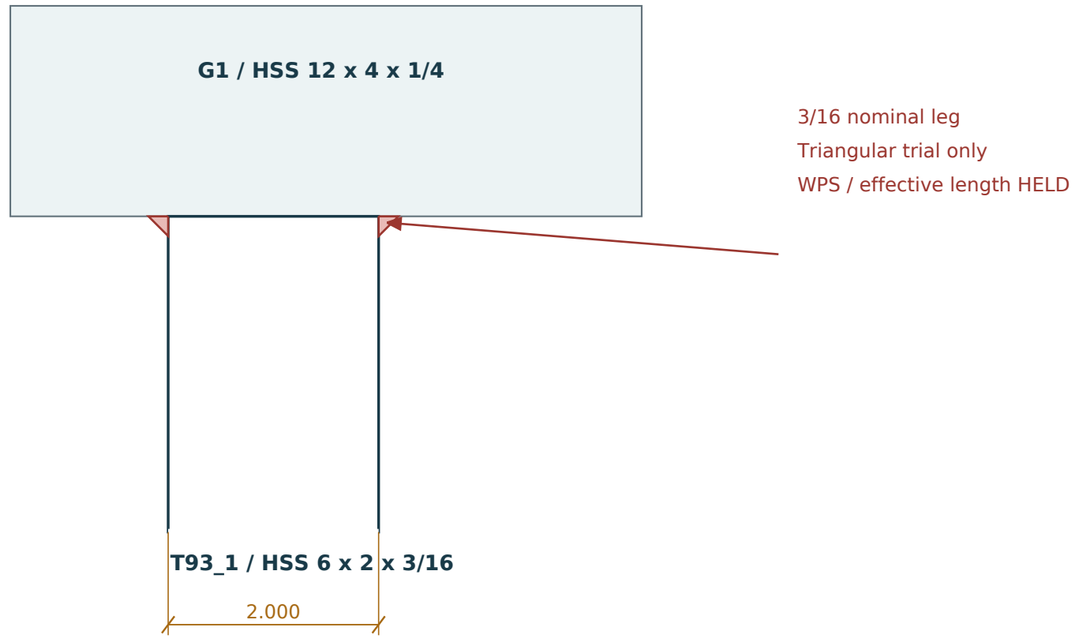


G1 receiver face continues below this detail

Branch T93_1: x92..94; y-32..32; z25.003937..31.003937.

Two side fillets only; no top weld across the floor-bearing plane.

I2.2 PLAN / x-y / WELD CROSS-SECTION ENLARGEMENT



Receiver inner face y = 32; branch cut envelope ends on that face.

Nominal contact plane is not an accepted root gap or saw allowance.

I2.3 DIMENSION AND MATERIAL CHANGES

Item	I-CM6 study
Central span count / length	16 x 64.000 in
Nominal section / design wall	6 x 2 x 3/16 / .174 in
Catalog I / S	10.5 in^4 / 3.50 in^3
Catalog unit weight	9.42 lb/ft
Material candidate	ASTM A500 Grade C

I2.4 LOCAL JOINT ACCEPTANCE REMAINS OPEN

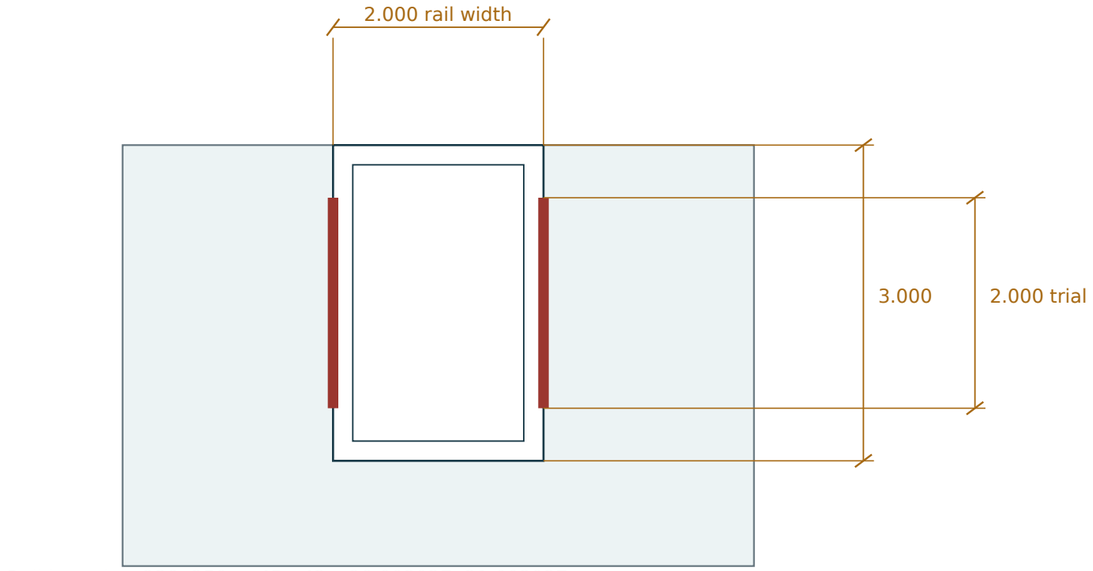
The deeper section spreads the trial weld group over 4.5 in. Main-girder face plastification, punching, local bending, load introduction, fatigue and weld procedure still control release. A weld-metal stress comparison alone does not establish joint capacity.

Trial end keepouts = 0.750 in at both branch ends in z. Actual product corners, weld seam, torch access and effective weld length must be checked on the supplied HSS.

I-SR32 / WIDER SEAT-RAIL END CONNECTION

Worked governing study: S-44_10 at its x238 start face on retained outer crossmember T237_0. This thin receiver is not strengthened by I-CM6.

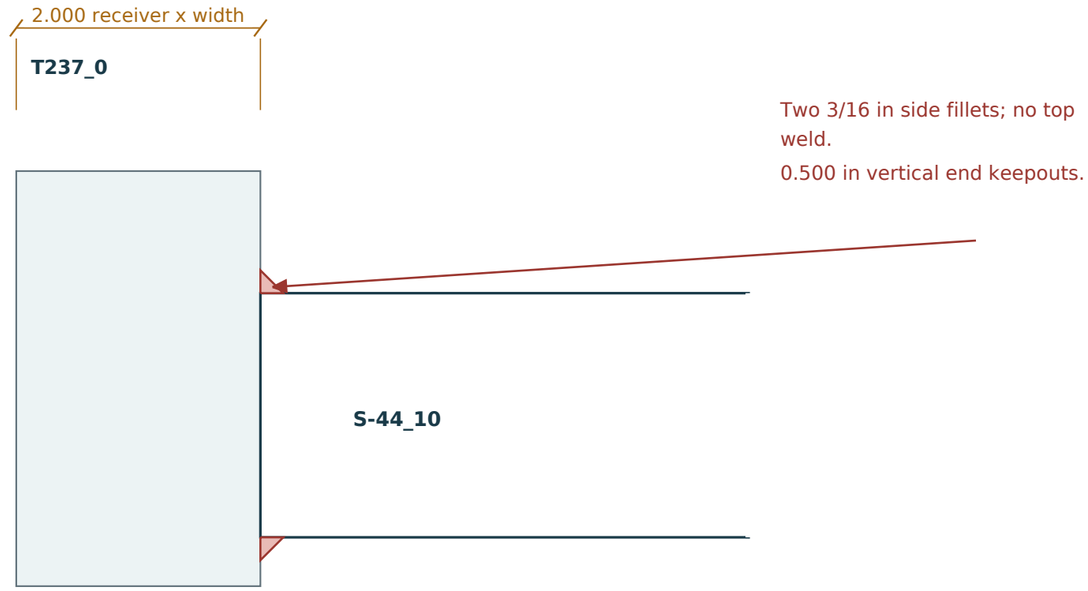
I3.1 END ELEVATION / y-z / AT x = 238



Rail bounds: x238..259; y-45..-43; z28.003937..31.003937.

Receiver T237_0: HSS 4 x 2 x 1/8; design wall .116 in.

I3.2 PLAN / ACTUAL CONTACT AND TRIAL FILLET SIDES



Rail length = 21.000 in; all 52 I-S32 rails retain previous end planes.

I3.3 COMBINED LOAD SENSITIVITY / ONE JOINT

Input / result	Study value
Vertical end force	2.809 kN
Axial-force bound	1.496 kN
End moment bound	0.258 kN m
Analysis case	S-44_10:start:LX+1:r1
Nominal rail / design wall	HSS 3x2x3/16 / .174 in

Material: A500 Grade C candidate. Sources I-F01-07. Actual local joint capacity = HELD.

I3.4 EFFECTIVE LENGTH IS A GOVERNING UNKNOWN

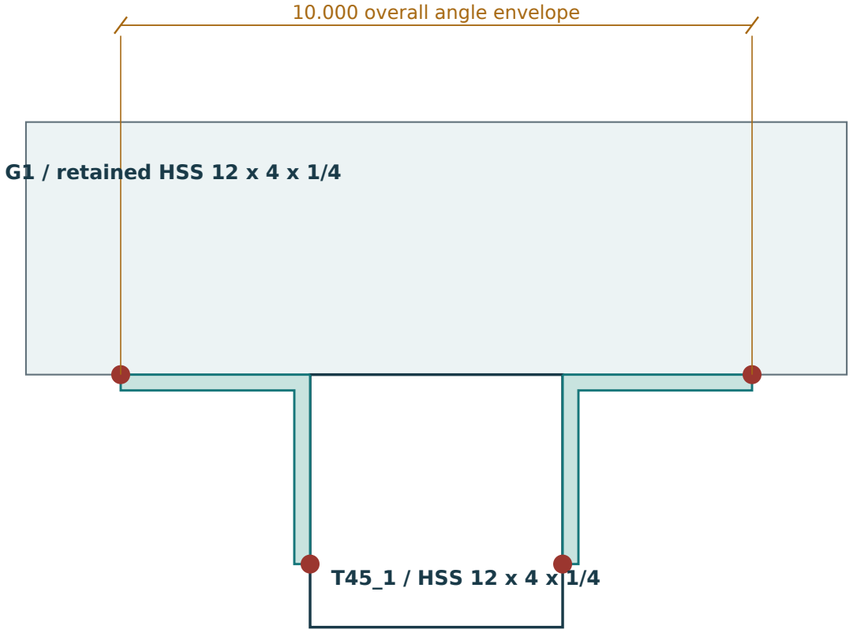
Assumed effective length / each side in	Weld-only ratio
2.000	0.431
1.500	0.757
1.000	1.680

Drawn 2.000 in is not automatically effective. Local receiver wall, actual toe/root stress and fatigue remain unaccepted. A ratio below 1 covers only this reference weld-metal screen.

I-BA / CENTRAL BOLSTER TO MAIN GIRDER

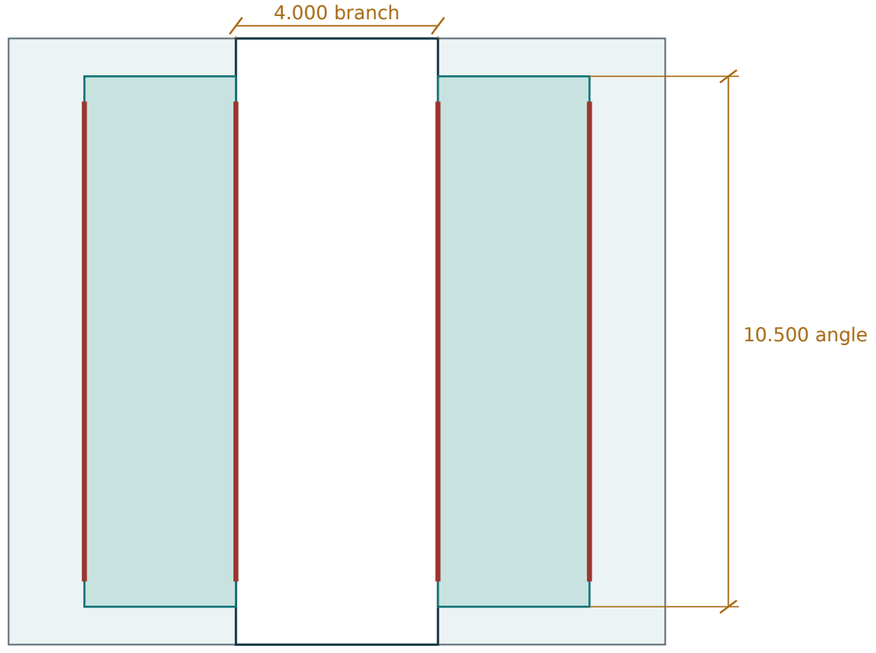
Two vertical L3 x 3 x 1/4 angles at each of four central bolster ends. This is a load-path development option; it is not a released moment connection.

I4.1 PLAN / x-y / FRONT RIGHT NODE



Red points: vertical toe-only weld paths (two per angle). No heel weld is assumed.

I4.2 ELEVATION / x-z / ANGLE HEIGHT



Angle z19.753937..30.253937; weld z20.253937..29.753937.

I4.3 COMPONENT AND WELD STUDY

8 angles total; 10.500 in nominal cut envelope. Candidate angle material: A36 minimum strength for screening; source listing grade conflict must be resolved by the mill certificate. Ideal sharp-corner geometry does not prove actual angle/HSS fit.

16 toe weld lines: 3/16 in trial leg x 9.500 in drawn length. Actual effective lengths, WPS, root clearance and local main/bolster walls remain held. No shared load credit to an original direct end weld.

I4.4 ROOT RESTRAINT CAN OVERTURN THE SCREEN

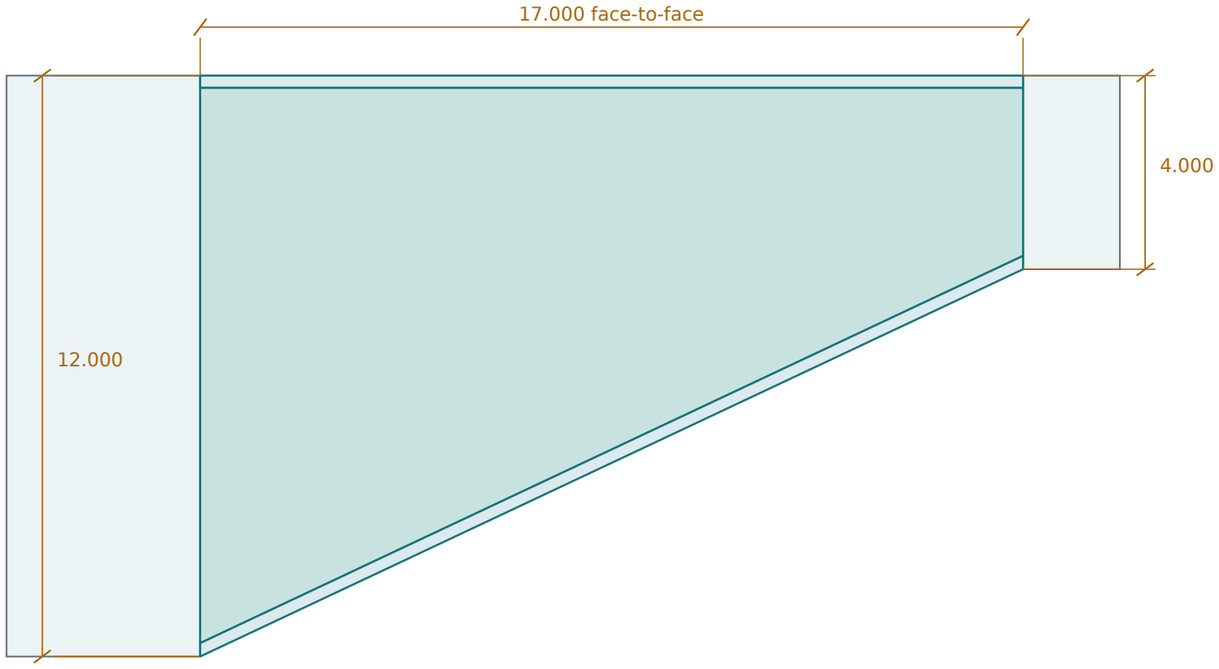
Root restraint fraction	Weld-only ratio
0.00	0.393
0.25	0.831
0.50	1.287
1.00	2.208

Balanced 2g sensitivity: 64.724 kN end shear; actual fixity and bogie support law unresolved.

I-BT / TAPERED OUTER BOLSTER BOX

Four proposed 17 in arms transition from 12 in depth at the main girder to 4 in at the perimeter. The shallow perimeter does not become a deep receiver.

I5.1 SIDE / y-z / FRONT RIGHT OUTER ARM



Top z31.003937 retained. Sloping underside rises 8.000 in over 17.000 in.

I5.2 PLATE BREAKDOWN / FINISHED STUDY ENVELOPES

Part suffix	Qty	Dimension / status
TOP	1	17.000 x 4.000 x .250
BOT	1	4.000 wide; .250 normal thick
W1 / W2	2	17.000 long; .250 thick
Root / tip depth	-	12.000 / 4.000 overall
Material	-	ASTM A572 Grade 50 candidate

Bottom plate length along slope = 18.788294 in before vertical end cuts / weld preparation. Web root height = 11.473702 in; tip height = 3.473702 in. Dimensions use non-overlapping nominal plate geometry.

Seam and end weld sizes, joint preparations, closing sequence, vent/drain details, end reinforcement and local receiving-wall resistance are HELD. Do not substitute a guessed all-around weld.

I5.3 END SECTIONS / PLATE THICKNESS



ROOT / y36

4.000 wide

.250 plate



TIP / y53

4.000 wide

.250 plate

I5.4 LOAD TRANSFER AND VALIDATION LIMIT

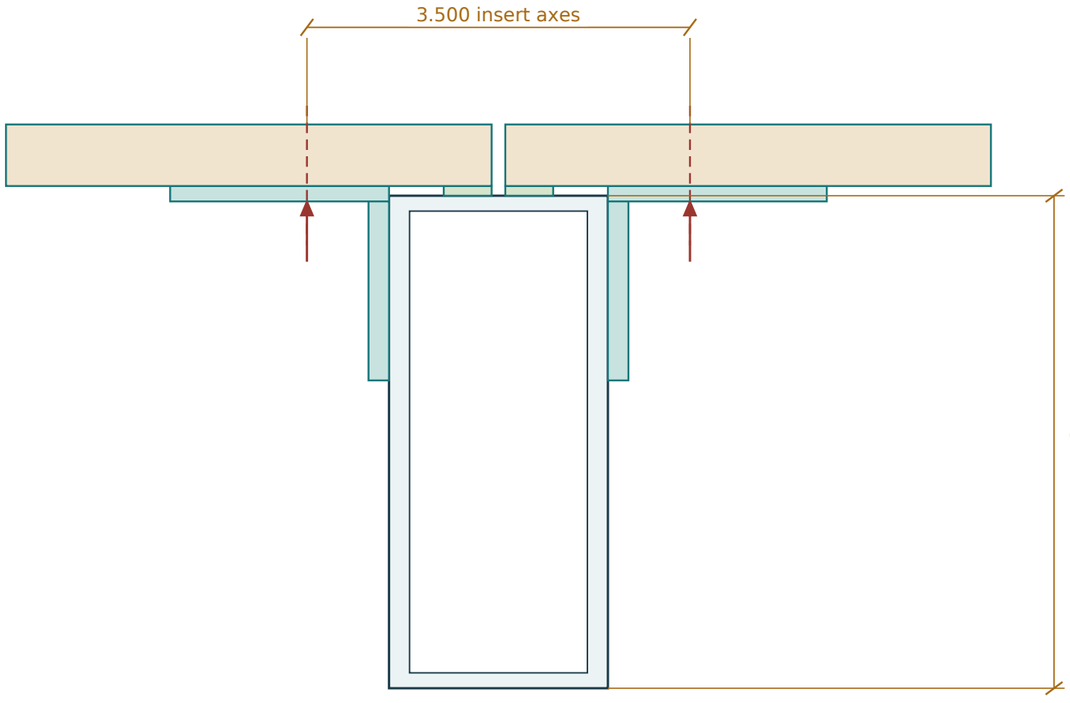
The tapered plates resolve the 12-to-4 in depth mismatch geometrically. Downward tip loads of 1 / 5 / 10 kN are analytical probes, not resolved body-wall reactions or rated capacity. Actual vertical, lateral, longitudinal and torsional demands remain open.

No bogie bearing, pivot, mount hole or support patch is assigned by this sheet.

I-FB / I-FR / FLOOR BEARING AND UNDERSIDE RETENTION

Representative ordinary transverse seam at x93 over I-CM6. Existing H-P125 finished-panel envelopes and 1/8 in internal gap are retained.

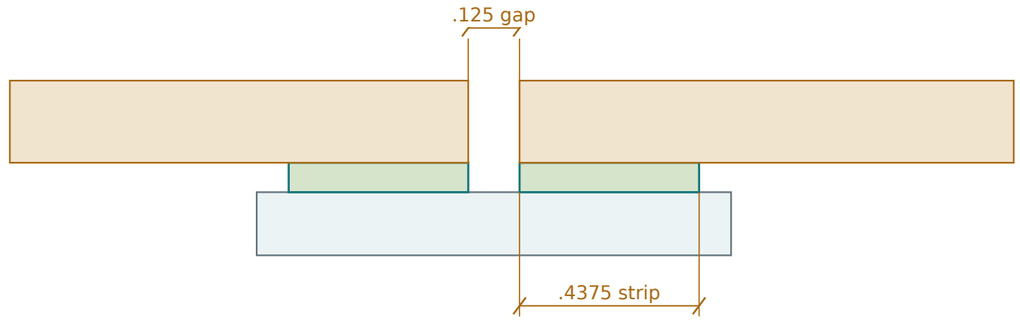
I6.1 SECTION / x-z / AT y = 0 / STUDY DETAIL



Arrows identify upward bolt axes; OEM insert dimensions and ratings are HELD.

Candidate clip bolts: 3/8-16 Grade 5; 7/16 in study flange holes.

I6.2 BEARING ENLARGEMENT / SEAM NORMAL



Two 7/16 in wide x 3 mm rigid bearing strips; 12 in demonstration length.

Nominal CM6 top flat = 1.156 in; .078 in outer margin each side.

NP510A glass-epoxy is a comparison candidate. Typical compression data are not allowable values. Minimum-thickness, fire/smoke, long-term load and environmental acceptance remain open.

I6.3 ONE PAIRED CLIP PROTOTYPE / A36 CANDIDATE

Feature	Prototype geometry
Flange / each	2.000 x 1.500 x .1875 in
Web / each	1.500 x 2.180610 x .1875 in
Bolt offset from HSS side	.750 in
Insert offset from panel edge	1.6875 in
Frame-side trial welds	2 x 1.500 long, 1/8 leg / clip
Web/flange trial weld	1 x 1.500 long, 1/8 leg / clip

Weld elbow moment, insert pullout, panel closeout, HSS wall and temperature movement require release. Two clips shown do not establish full-floor attachment.

I6.4 CONTINUOUS LOAD PATH / HELD SUPPLIER DESIGN

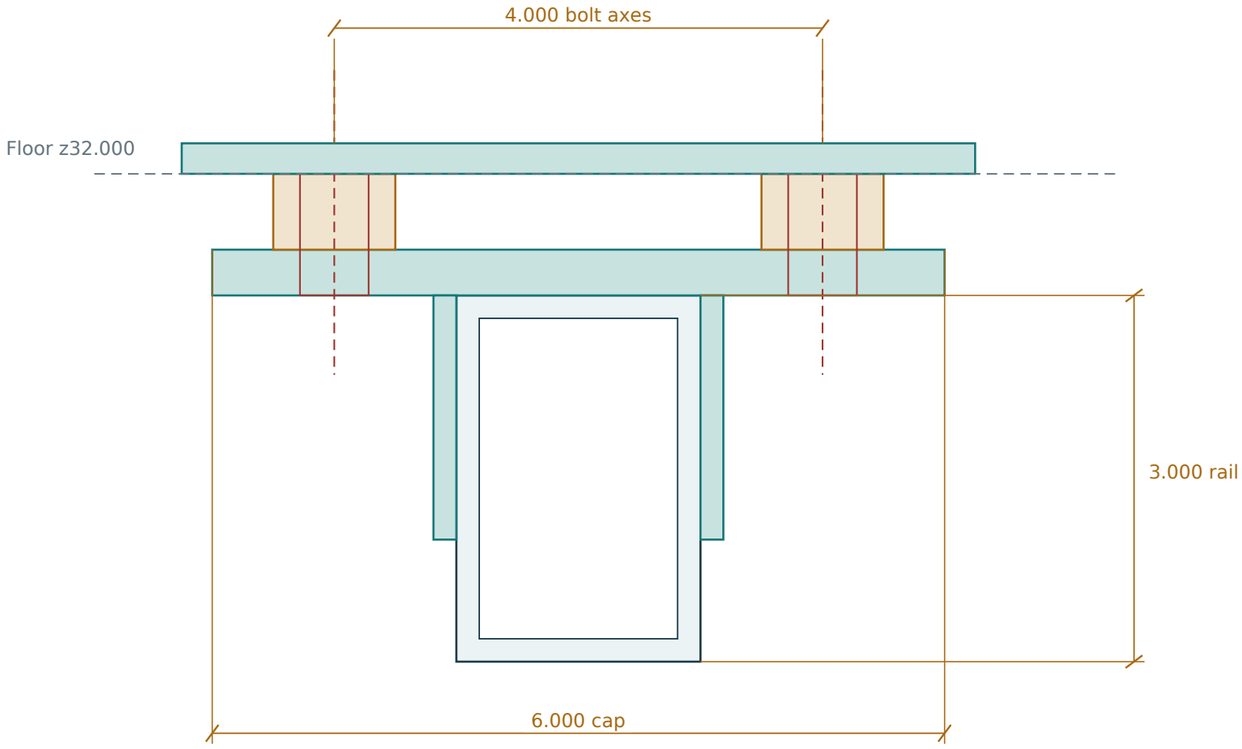
Panel factory closeout and insert must deliver load into the bonded panel structure. The 1 in closed-edge envelope is a reservation, not a fabricated insert. Skin screws and compressing honeycomb are not assigned structural capacity.

Clip-to-insert electrical barrier, actual thickness and finish must be coordinated. Adjust clip height to the accepted barrier; final spacing and complete-car quantity remain HELD.

I-SH1 / SEAT HARDPOINT AND RIGID FLOOR BYPASS

Example foot at x97 / y20 (HB03R). Candidate geometry preserves the H-S16 16 x 24 in foot pattern and does not clamp the honeycomb floor.

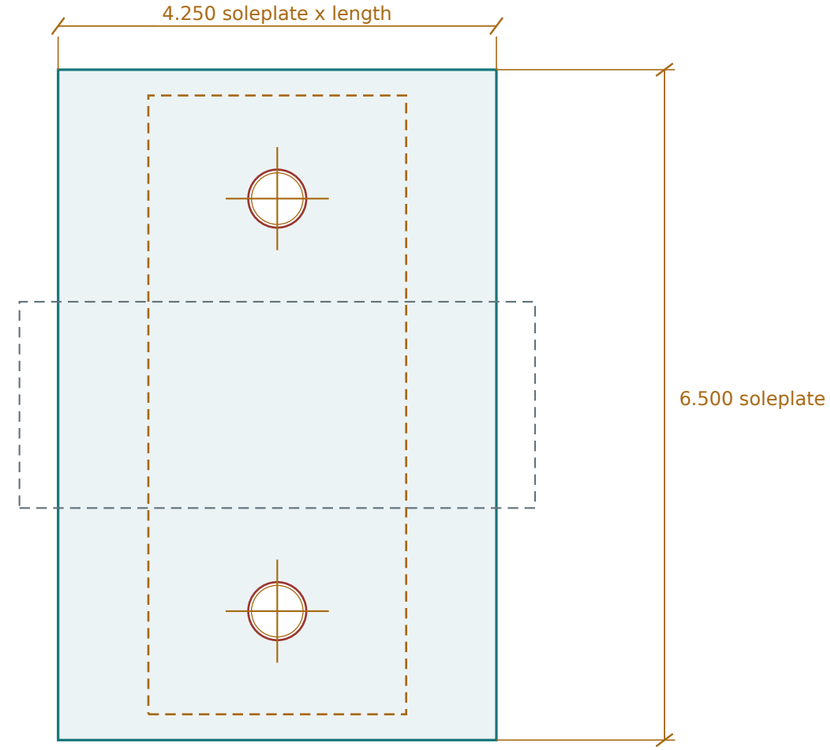
I7.1 TRANSVERSE SECTION / y-z / ONE FOOT



Spacers: 1.000 OD; 9/16 bore; .621063 long (datum difference).

2.750 x 6.250 in finished floor-opening envelope; OEM closure / seal HELD.

I7.2 PLAN / BOLT AND PLATE COORDINATES



Local origin: selected H-S16 foot centre. Holes at x0 / y +/-2.000; 9/16 in diameter.

Cap 2.500 x 6.000 x .375; soleplate 4.250 x 6.500 x .250.

I7.3 PHYSICAL CONNECTION / ASSEMBLY ORDER

Rail reinforcement first: A36 cap and two 2.500 x .1875 x 2.000 cheek plates. Then fit and seal the separate rigid spacers through the floor. A36 soleplate bears on spacers. Two outboard 1/2-13 Grade 5 bolts are candidates; length, nut/washer assembly and torque are held.

Seat frame-to-soleplate attachment remains unassigned.

I7.4 CALCULATED GEOMETRY / ACCEPTANCE HELD

Bolt, plate bending, rail wall, local load introduction, welds, prying, fatigue, torque and retained clamp force require coordinated release. Spacer length is a nominal stack calculation; finish, actual flatness and seat supplier base dimensions remain unresolved.

Trial welds: cheek/rail 4 x 1.5 in at 1/8 leg; cap/cheek 2 x 2.5 in at 3/16 leg. WPS and capacity HELD.

PULSIPHER FARMS / CHRISTMAS RAILWAY

ASSEMBLY CONTROL / ENGINEERING RELEASE RECORD

Use this sheet to review the proposed sequence and record the missing acceptance evidence. No fixture, weld, torque or tolerance is released.

I8.1 CONNECTION DEVELOPMENT SEQUENCE

01 Freeze demand set

Responsible vehicle engineer: accepted mass / load combinations / fatigue / dynamics, assigned to each local joint.

02 Return component data

Steel supplier, bogie designer and floor/seat supplier: actual section corners, material traceability, support and attachment data.

03 Release shop detail

Connection engineer: local shell checks, plate/fastener/weld design, access, fit-up, coating interfaces and tolerances.

04 Trial-fit first article

Fabricator: measure A/B/C; fit reinforcement before concealment; verify clamp/torch/tool access; record actual clearances.

05 Execute accepted procedure

Qualified fabricator: approved WPS / personnel / sequence; record heat, weld maps, distortion and required inspections.

06 Verify connected assembly

Engineer-directed joint/seat/panel tests and dimensional survey; no occupied trials before release.

07 Disposition and duplicate

Resolve deviations; sign component, assembly and vehicle gates before production repetition.

I8.2 RECORD FIELDS / NO DEFAULT ACCEPTANCE VALUES

Record	Requirement	Actual / release
Joint / component ID	_____	_____
Material / heat / certificate	_____	_____
Shop detail / revision	_____	_____
Measured wall / corners	_____	_____
Cut / fit-up / orientation	_____	_____
WPS / welder ID	_____	_____
Inspection / NDT plan	_____	_____
Tool and isolation clearance	_____	_____
Final plane / distortion	_____	_____
Test method / result	_____	_____
Deviation disposition	_____	_____
Responsible release	_____	_____

Farm crew can label, inventory, record measurements, prepare safe handling and perform released finish work. Connection design, fixture acceptance, critical welding, NDT and structural approval require the appropriate specialists.

Steel purchase and production cutting wait for the connection revision, actual supplier dimensions and released material/WPS/inspection requirements. No bolt, weld leg or plate shown here overrides an unresolved hold.

VERIFIED / source data CALCULATED / stated inputs STUDY / proposed geometry HELD / required before release